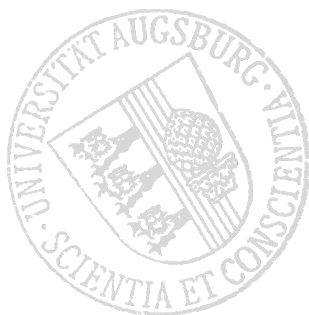


Modeling and Simulation of Biological Processes
University of Augsburg

Master Thesis

**How Reliable Are Molecular Predictions?
From Uncertainty Quantification to Active
Learning in Cheminformatics**



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Abbreviations

AL	Active learning
AULC	Area under the learning curve
BI	Bregman information
GBDT	Gradient boosting on decision trees
GNN	Graph neural network
MCC	Matthews correlation coefficient
MLP	Multi-layer perceptron
NIBR	Novartis Institutes for Biomedical Research
NN	Neural network
NODE	Neural oblivious decision ensembles
QSAR	Quantitative structure-activity relationship
SIMPD	Simulated medicinal chemistry project data
SMILES	Simplified molecular input line entry System
TabPFN	Tabular prior-data fitted network
TS	Temperature scaling
UQ	Uncertainty quantification

Abstract

Drug discovery increasingly relies on machine learning models to guide costly experimental decisions, making reliable estimates of model uncertainty absolutely essential. This thesis systematically investigates multiple uncertainty quantification methods for machine learning-based bioactivity prediction. Calibration quality and different uncertainty characteristics were evaluated along with classic predictive performance across several molecular datasets, classification models and feature representation types. Furthermore, uncertainty estimates were integrated into iterative active learning workflows to assess their effectiveness in guiding compound selection under varying experimental budgets.

The results indicate that calibration quality depends strongly on the chosen model architecture or molecular representation, unlike predictive performance, which is stable across different settings. Across all investigated models, miscalibration was observed with a tendency towards overconfident predictions, most prominently for neural network models. Predictive entropy and ensemble disagreement provided complementary uncertainty estimates, capturing total and epistemic uncertainty, respectively. In active learning experiments, uncertainty-based acquisition strategies frequently outperformed random sampling, emphasizing the importance of calibrated predictions for efficient data acquisition. Combining uncertainty with diversity-based selection yielded the most consistent improvements.

Overall, the findings demonstrate that uncertainty-aware machine learning can improve decision reliability and data efficiency in molecular prediction. This work contributes to a better understanding of predictive uncertainty in cheminformatics and supports the development of more reliable machine learning-driven drug discovery pipelines.

1 Introduction

Machine learning (ML) is increasingly used in chemistry to guide safety-critical decisions in expensive and time-consuming molecular experiments. This introduces the need for not only correct, but also trustworthy predictions. Therefore, ML models must communicate their uncertainty alongside of providing accurate predictions, allowing practitioners to assess if these predictions can be trusted. Reliable uncertainty quantification (UQ) enables informed confidence in predictions, while efficient learning strategies aim to minimize experimental effort where labeling is costly. Together, these challenges define a central research direction at the intersection of ML and cheminformatics.

1.1 Machine learning in cheminformatics

The prediction of molecular properties from chemical structure has long been a key objective in cheminformatics. Quantitative Structure-Activity Relationship (QSAR) modeling provides a foundational framework for this task by establishing statistical relationships between molecular descriptors and biological or physico-chemical activity. Traditionally based on linear statistical models and handcrafted molecular features, QSAR approaches have evolved substantially with the emergence of modern ML methods [1]. Contemporary ML models extend the QSAR paradigm by enabling flexible non-linear modeling and automated representation learning, allowing increasingly accurate prediction of molecular properties across diverse domains.

The application of ML in cheminformatics now ranges from protein design, over materials and drug discovery, to molecular property prediction and further use cases [2]. In this domain, ML models aim to learn relationships between molecular structure and experimentally observable properties, enabling rapid in silico screening prior to costly laboratory experiments. A typical ML workflow comprises data collection and curation, molecular representation through feature extraction or learned embeddings, model selection, and subsequent model training and evaluation.

The adoption of ML offers several advantages for chemical research. Predictive models can reduce experimental costs and enable the exploration of vast chemical spaces that would be infeasible to investigate experimentally, accelerating chemical innovation [3, 4]. However, these benefits are accompanied by important limitations. Model performance strongly depends on data quality and

coverage, and predictions may become unreliable when applied to molecules that differ from the training distribution, also known as out-of-distribution problem [5]. Consequently, assessing prediction reliability remains a principal challenge in practical applications.

Depending on the prediction target, molecular ML tasks can be formulated either as regression or classification problems. Regression models are commonly used to predict continuous physicochemical properties such as solubility, binding affinity, or toxicity measurements [6]. In contrast, classification settings arise when compounds are assigned to discrete categories. A particularly important example is the prediction of compound bioactivity against biological targets, which is frequently formulated as a binary classification task distinguishing active from inactive molecules [7]. Such bioactivity prediction problems constitute a main application scenario in drug discovery and form the basis for many downstream decision-making processes.

1.2 Uncertainty quantification for safety-critical applications

While modern models can achieve high predictive performance in tasks like bioactivity prediction, accurate predictions alone are insufficient for reliable decision-making, motivating the need for principled uncertainty quantification methods. Predictive uncertainty aims to quantify the confidence associated with model predictions. Commonly, a distinction is made between aleatoric uncertainty, arising from inherent data noise, and epistemic uncertainty, reflecting limited model knowledge due to insufficient training data [8]. Only the latter is considered to be reducible [9].

Quantifying both types of uncertainty is especially important in safety-critical domains. Drug discovery applications within cheminformatics define high safety requirements, as prediction errors can directly influence compound selection decisions with downstream consequences for patient safety, experimental resource allocation, and regulatory compliance regarding false positive as well as false negative predictions [10]. Reliable uncertainty estimates therefore enable informed decision-making, allowing practitioners to prioritize experiments, detect unreliable predictions, and mitigate risks associated with model deployment.

1.3 Efficient labeling through active learning

In downstream applications, improved uncertainty estimates can be leveraged to guide active learning (AL). AL aims to increase data efficiency by iteratively selecting the most informative samples for labeling rather than relying on randomly collected training data, which is most beneficial when labeling cost is high. In a typical AL workflow, a model is first trained on a small initial labeled dataset and subsequently used to evaluate a pool of unlabeled candidates. A query method then selects a subset of samples whose acquisition is expected to provide the greatest improvement to the model. The selected samples are labeled by an annotator and incorporated into the training set for the next model training cycle, and this process is repeated over multiple iterations (see Figure 1), with the aim to improve model performance optimally.

The effectiveness of AL depends on several factors, including the size of the initial labeling budget, the underlying predictive model, the choice of query method, the number of samples acquired per iteration (query size) and the total number of acquisition cycles.

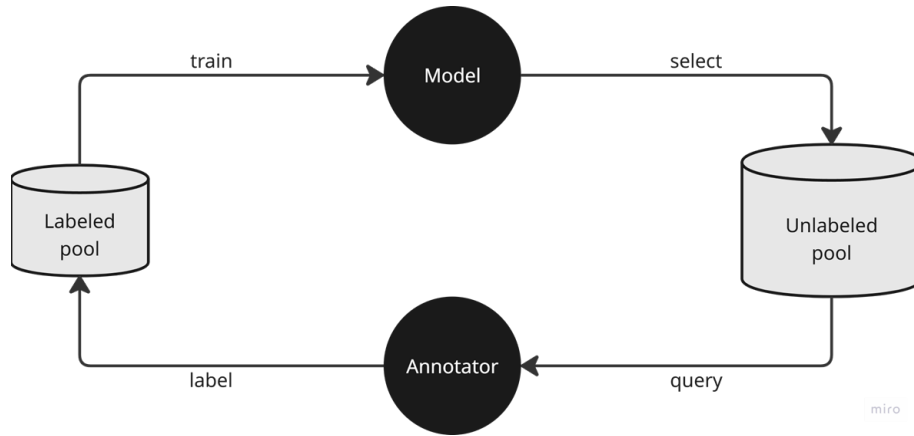


Figure 1: Active learning cycle

The labeled pool is extended by labeling samples from the unlabeled pool in multiple iterations.

A central challenge in AL lies in balancing exploration and exploitation. Exploration aims to acquire diverse or previously unseen regions of the input space to improve general model knowledge, whereas exploitation focuses on refining decision boundaries in regions already identified as promising [11]. Uncertainty-based query strategies naturally address this trade-off by prioritizing samples for which the model lacks confidence. In particular, epistemic uncertainty is especially relevant for AL, as it highlights regions where additional observations are

expected to most effectively reduce model uncertainty. Consequently, reliable uncertainty quantification plays a crucial role in enabling effective AL for molecular machine learning applications.

1.4 Research objectives and contributions

In current research, there is no systematic analysis of UQ for multiple molecular datasets and their usage across several model architectures and feature representations. Furthermore, it remains an open question how uncertainty-based active learning can be applied for bioactivity prediction. Consequently, this thesis aims to answer the following research questions: (1) How do model architecture and dataset representation affect reliability of molecular predictions, and (2) can uncertainty estimation improve AL with molecular data?

The main contributions of this work include a systematic comparison of multiple UQ methods for molecular bioactivity prediction, i.e. a comprehensive evaluation of model calibration using established metrics, and an experimental framework enabling consistent evaluation across datasets, models, representations and uncertainty measures. Furthermore, this work yields practical insights into the applicability of uncertainty-based methods in AL for data-efficient cheminformatics. This work focuses on empirical evaluation rather than the development of novel uncertainty estimation algorithms.

With these contributions, this work seeks to provide both methodological insights and practical guidelines for developing reliable ML models under limited and uncertain data conditions. Thereby, experimental costs in fields like drug discovery are aimed to be reduced in the long term by providing trustworthy predictions and efficient data acquisition techniques via AL.

2 Related work

This chapter reviews existing research related to machine learning for molecular bioactivity prediction, uncertainty quantification, and active learning. Building on the conceptual background introduced in chapter 1, the following sections position the present work within prior studies, highlighting methodological advances, evaluation practices, and remaining challenges addressed in this thesis.

2.1 Molecular prediction with machine learning methods

Machine learning models can achieve strong predictive performance for bioactivity prediction among other molecular prediction tasks across many benchmark datasets, as Wu et al. [12] have shown with MoleculeNet, a benchmarking platform using a variety of curated datasets. Concerning different model architectures, some prior works present superior performance of deep learning methods [13, 14], while others highlight the robustness and efficiency of gradient boosting models [15, 16]. Representation choice, ranging from 2-dimensional molecular fingerprints to learned embeddings, strongly influences predictive accuracy as well [17].

However, reported performance improvements are often inconsistent across datasets and studies, indicating sensitivity to data distribution, preprocessing and other study-specific dependencies. Most studies primarily focus on predictive accuracy, with limited attention to uncertainty. This highlights the need for a systematic reliability analysis of predictions across models, datasets and representations.

2.2 Uncertainty quantification in bioactivity prediction

Other prior studies have focused on uncertainty in cheminformatics already. Hirschfeld et al. [18] evaluated several methods on five regression datasets, finding that none of the methods was superior across all cases. The authors only investigated neural network models, without comparing multiple model architectures. A larger-scale study evaluated ensemble methods across 32 cheminformatics datasets of varying sizes and difficulty, testing five modeling techniques with four molecular featurizations [19]. This study also focused on regression tasks, with lacking attention to classification models. Yu et al. [20] tested a hybrid framework combining distance-based and Bayesian approaches on 24 bioactivity datasets, demonstrating improved uncertainty ranking ability, but

stating that the most appropriate UQ method is dependent on the downstream task.

Dai et al. [21] identified calibration as a key performance metric alongside accuracy, precision, and efficiency for evaluating uncertainty quantification methods, noting that recalibration techniques can improve poorly calibrated models. Consequently, this work focuses on calibration quality as an uncertainty measure, which most studies disregarded so far.

The field has explored both traditional approaches such as empirical, frequentist, and conventional Bayesian methods [10], and modern deep learning-based techniques, including explainable UQ methods that attribute uncertainties to molecular atoms [22]. However, UQ in computational chemistry lacks common standards for reporting and validating prediction uncertainty, indicating the field remains in early stages despite growing attention.

2.3 Active learning for molecular discovery

AL has been studied in cheminformatics as an iterative feedback process to efficiently identify valuable data in chemical space, with applications reviewed across drug discovery stages including compound-target interaction prediction, virtual screening, molecular generation, and molecular properties prediction [23, 24]. Uncertainty-based acquisition consistently emerges as the most effective across studies, with A. Polash et al. [25] reporting that uncertainty-based querying achieved a Matthews Correlation Coefficient (MCC) of 0.90 with 20% of training data, significantly outperforming greedy (MCC of 0.30) and random selection (MCC of 0.70) at the same data volume. Therefore, the success of AL experiments relies on high quality uncertainty estimates.

In summary, ML models have been widely applied, but reliability keeps being underexplored. Different UQ methods have been proposed, but current research lacks a systematic comparison of methods in bioactivity classification tasks. Furthermore, calibration quality and post-hoc calibration methods are rarely analyzed comprehensively. AL is a promising approach to reduce high data demand in cheminformatics, but it has not clearly been linked to UQ quality directly. This thesis addresses these gaps through a unified experimental framework evaluating UQ methods and uncertainty-driven active learning across multiple datasets, models, and molecular representations in the field of molecular bioactivity prediction.

3 Data and methods

This thesis makes use of three datasets for analyzing and comparing the uncertainty behavior of multiple ML models, working with the following setup.

3.1 Dataset description and analysis

The datasets used in this study were created by Landrum et al. [26] for the purpose of validating ML models in the context of medicinal chemistry projects and are publicly available. They were obtained from the author’s GitHub repository¹, versioned by the latest release from December 10, 2023. The authors provide 99 public datasets in total, of which three were selected for this study considering the size of the datasets and other criteria presented in the following.

All three datasets contain molecular compounds, represented by the Simplified Molecular Input Line Entry System (SMILES) strings. SMILES is a line notation of chemical structures that is able to characterize the atoms, bonds, branches and cyclic structures of compounds using human-readable and machine-interpretable ASCII (American Standard Code for Information Interchange) symbols [27]. Each compound is a sample with an assigned target label (0 or 1) that refers to the bioactivity of the compound against a specific biological target protein. The biological target is fixed across all samples of one dataset. The characteristics of target proteins are shown in Table 1. The datasets will be further referred to by the abbreviation of the respective target, CA-IV, hERG and OX1R.

Dataset	Target name	Target class	Target relevance
CA-IV	Carbonic anhydrase IV	Lyase	Gas exchange, pH regulation
hERG	Human ether-a-go-go related gene	Ion channel	Cardiac function
OX1R	Orexin receptor 1	Membrane receptor	Sleep-wake-cycle

Table 1: Target characteristics of selected datasets

Each target refers to the protein that bioactivity of compounds in the dataset is tested against.

The datasets have been constructed using the simulated medicinal chemistry project data (SIMPDP) algorithm [26], which splits datasets into early and late compounds in a time-based manner to mimic differences in datasets from

¹https://github.com/rinikerlab/molecular_time_series

real-world medicinal chemistry projects. These differences reflect the order in which the compounds were made or tested in the project, since late compounds are often designed or selected on the basis of knowledge derived from early compounds [26]. Therefore, each compound in the datasets is assigned to the train or test set in a 80:20 ratio already, which means that it was categorized as an early or late sample by the SIMPD algorithm, respectively.

Based on experiments with medicinal chemistry project data from the Novartis Institutes for Biomedical Research (NIBR) that was used to train the SIMPD algorithm, public ChEMBL datasets from project-specific assays with measured K_i values were curated from ChEMBL32 [28]. The K_i value represents the inhibition constant and measures binding affinity of a compound to a biological target. The lower the K_i value, the greater the ability of the respective compound to bind to receptors or enzymes, which means that a lower K_i value is associated with a higher pharmacological potency. The value can be determined experimentally [29]. Service assays were excluded from the data collection. To also eliminate large screening assays, the datasets were restricted to 300 to 1,000 measurements for unique compounds for any biological target. A biological target in bioactivity prediction is defined as a protein–organism pair, since homologous proteins from different species exhibit sequence and structural variations that can substantially alter ligand binding and measured activity. In the data collection, different organisms were included. However, in this study, all selected datasets consider human targets.

For the label assignment, differentiating samples between the active and inactive class, the authors split each dataset into 40% active and 60% inactive samples [26]. This results in an activity threshold for each dataset, given as a pChEMBL value, which can be calculated as a negative logarithmic measure of the K_i value as follows:

$$\text{pChEMBL} = -\log_{10}(K_i) \quad (1)$$

For this study, three datasets with a pChEMBL threshold between 6.0 and 6.5 have been selected to provide reasonable binary class assignments, as suggested by Lenselink et al. [30]. All compounds in each dataset have been salt-stripped and neutralized using the RDKit’s ChargeParent standardizer.

The properties of the three selected datasets are listed in Table 2.

Dataset	Filename	pChEMBL threshold	# Compounds
CA-IV	CHEMBL3729-1.csv	6,0	364
hERG	CHEMBL240-2.csv	6,1	579
OX1R	CHEMBL5113-1.csv	6,1	678

Table 2: Properties of selected datasets

The datasets are split upon the pChEMBL threshold into active and inactive compounds. The filenames correspond to the filename in the source repository².

Chemical Data Analysis

Prior to model training, the dataset was characterized with respect to its structural diversity and scaffold composition. Such analysis provides insight into the coverage of chemical space and informs the expected difficulty of generalization beyond known molecular series.

Structural diversity was assessed using pairwise Tanimoto similarity computed from Morgan fingerprints. The Tanimoto similarity is an association coefficient, developed for comparison of binary vectors, which considers the commonly set bits in the fingerprints of both molecules [31]. The resulting similarity distribution (see Figure 2) indicates moderate redundancy, with most compounds exhibiting similarity values below 0.4 for all datasets, suggesting broad coverage of chemical space.

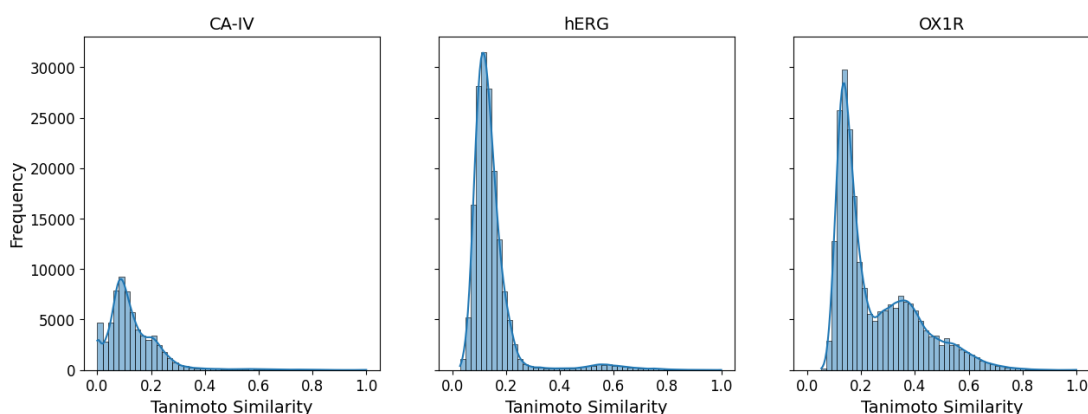


Figure 2: Pairwise Tanimoto similarity distribution

Pairwise Tanimoto similarities of compounds based on Morgan fingerprints are shown per dataset.

Next, unique scaffolds per dataset were counted. Molecular scaffolds represent

²https://github.com/rinikerlab/molecular_time_series

the core structural backbone of molecules and provide a chemically meaningful notion of similarity beyond fingerprint-based metrics. For the analysis, Bemis-Murcko scaffolds [32] were computed from SMILES and aggregated per dataset and dataset split (train vs. test set). Results are displayed in Table 3.

Dataset	Unique scaffolds (total)	Unique scaffolds (train)	Unique scaffolds (test)	Novel scaffolds in test (%)
CA-IV	130	112	44	40.9
hERG	336	283	98	54.1
OX1R	285	252	75	44.0

Table 3: Unique scaffold counts

Counts of unique Bemis-Murcko scaffolds per dataset and dataset split as well as the percentage of novel scaffolds in the test sets, compared to the respective train sets, is shown.

Based on this analysis, the five most common scaffolds are determined per dataset (see Figure 3), providing insight into dominant structural motifs within each dataset. The absence of strongly dominant scaffolds in CA-IV and hERG suggests relatively uniform chemical coverage, potentially supporting better generalization. For OX1R, a limited number of core structures dominate the dataset. This indicates a degree of chemotype imbalance, suggesting that a substantial portion of the data consists of closely related molecular series.

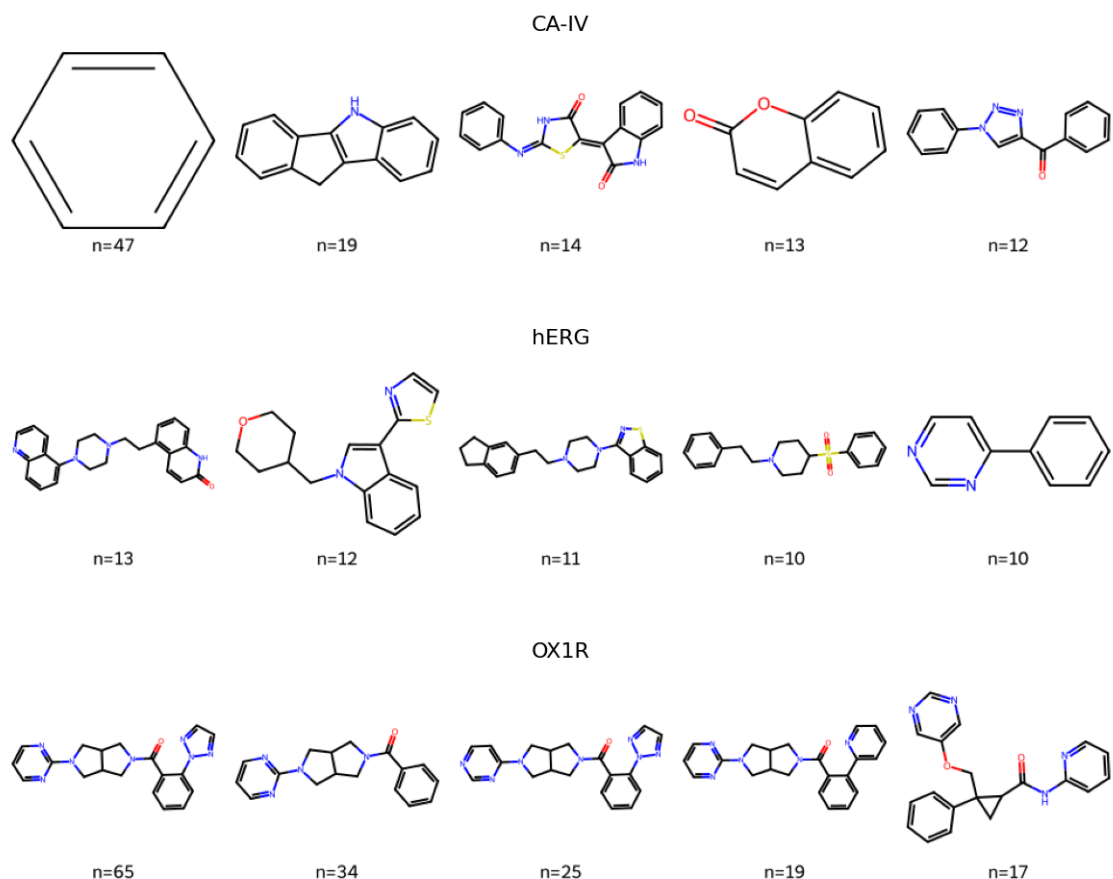


Figure 3: Most common scaffold structures

Per dataset, the shown molecular scaffolds occur most commonly with a frequency of n .

While such redundancy may facilitate model learning, it also increases the risk of overly optimistic performance estimates. Despite this structural concentration, the subsequent experiments focus on evaluating model behavior and uncertainty estimation performance, independent of detailed scaffold-level effects.

3.2 Uncertainty quantification pipeline

All datasets (CA-IV, hERG, OX1R) were used to train different ML models and quantify their uncertainty with several methods. The choice of models aims to cover a variety of model architectures currently present in state-of-the-art ML.

Models

As a baseline, a neural network (NN) was implemented in PyTorch³ based on the architecture used by Friesacher et al. [33]. It is a lightweight multi-layer

³<https://github.com/pytorch/pytorch>

perceptron (MLP) with two linear layers, with a ReLU (Rectified Linear Unit) and a dropout layer between them. A sigmoid function was applied to obtain class probabilities from the logits predicted by the last linear layer. Several hyperparameters were tuned using cross-validation, being the size of the hidden layer and the dropout rate for the dropout layer, as well as the weight decay and the learning rate for the optimizer.

The second model used in this study is CatBoost [34], which is an open-source gradient boosting toolkit. Gradient boosting is an ensemble method that builds a strong model by sequentially adding weak base learners, where each new learner is trained to correct the error of the previous ones by performing gradient descent in the functional space. CatBoost belongs to the family of gradient boosting decision trees (GBDTs), that use decision trees as the weak learner [35]. However, the base predictors are oblivious decision trees here, meaning that at the same depth level of a tree, all nodes apply the same splitting criterion. The CatBoost algorithm implements ordered boosting, which is a permutation-driven alternative to the classic gradient boosting [34].

The third model for evaluation of uncertainty quantification methods is TabPFN (Tabular Prior-Data Fitted Network) [36, 37], which is a transformer-based foundation model for tabular data. The model is pretrained on millions of synthetic datasets representing different prediction tasks generated by an approach based on structural causal models. TabPFN is able to perform inference directly on small unseen datasets. It can handle missing values automatically, encodes categorical variables and normalizes features.

Furthermore, a Neural Oblivious Decision Ensembles (NODE) model [38] was tested in this study. The architecture consists of multiple densely connected layers of differentiable oblivious decision trees that are trained end-to-end by backpropagation. In the end, the model was excluded from the study considering limitations in performance and runtime.

In addition to the single models, an ensemble model was constructed for each model type to access ensembling-based uncertainty quantification metrics. An ensemble model consists of multiple members. Each member is a single model that is trained on the training data independently and predicts on the test data. In the end, predicted probabilities are averaged across members to obtain a single prediction. Class predictions are derived from this mean probability. In order to increase training variability across members, each member is initialized with a

different random seed. Furthermore, bootstrap aggregation (bagging) was performed. This means that each member is fitted on a different sample of the training data. The corresponding training sample is of same size as the training data itself, resulting from resampling with replacement. Here, a fixed number of five ensemble members per ensemble model was selected, as suggested by Ovadia et al. [39].

Data preparation

Before training the models on the three datasets, the data was split into train, validation and test sets. Then, compounds are converted to different feature representations based on the SMILES string. Dependent on model architecture, calculated features were preprocessed further.

For the train-test split, the simulated time-based split from the dataset source was utilized, which is robust and respects the "continuity of design" [26] of lead-optimization datasets in drug design. The given split ratio is 80-20 for train vs. test. As post-hoc calibration methods were applied in the workflow, a separate validation set was needed that was neither used for training of the models nor had to be hold out for evaluation. Regarding that the train set sizes of the datasets were already small ($n < 550$), the calibration set for post-hoc calibration was split by 50% from the predefined test set in a randomized manner, stratified on the class label. For hyperparameter tuning with n -fold cross-validation, the train set was split into three folds. To take possible similarities of compounds into account, the samples were assigned to folds based on their membership regarding Butina clusters.

Butina clustering [40] is a distance-based clustering algorithm widely used in cheminformatics to cluster molecules based on structural similarities. Therefore, SMILES representations are translated into RDKit fingerprints with a maximum path length of five bonds. Then, the similarity between every two compounds is computed using the Tanimoto coefficient. This serves as a basis for constructing the distance matrix of the fingerprints with

$$\text{distance} = 1 - \text{similarity}. \quad (2)$$

Compounds are then clustered together, if they have a maximum distance below a specified cut-off from the cluster centroid. This cut-off is set to 0.3 here in order to obtain equal sized folds in the end. Based on their cluster assignment,

compounds are finally grouped into folds using GroupKFold from scikit-learn⁴.

The resulting complete data split is visualized in Figure 4.

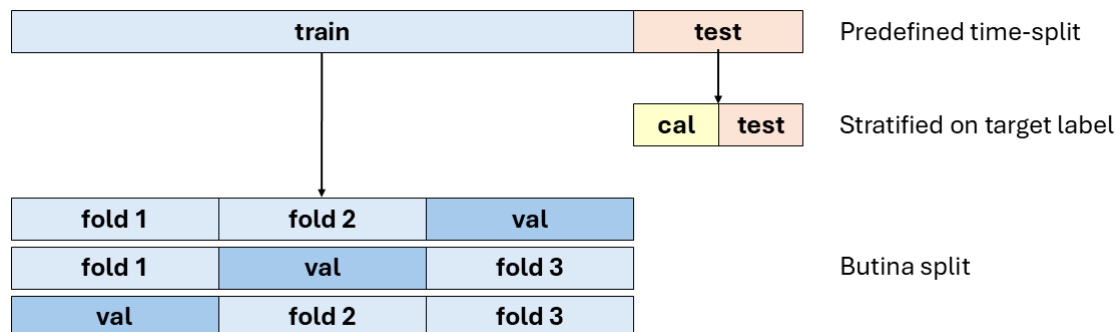


Figure 4: Dataset split

Each dataset is split into the training set (train), calibration set (cal) and test set (test). The training set is further split into three folds for cross-validation, where in each iteration, one fold is left out for validation (val).

In the next step, the compound representation was transformed from SMILES strings to different features. First, **RDKit physicochemical descriptors** from the cheminformatics toolkit RDKit⁵ were used. These are numerical tabular features describing the physicochemical properties of a compound, e.g. molecular weight or the number of radical electrons. There are 217 features computed for each compound.

The second feature set used in this study are **Morgan fingerprints**, which belong to the family of circular fingerprints [41]. Morgan fingerprints are created by applying the Morgan algorithm [42] to encode the atomic environment up to a radius specified by the user, considering atom invariants. Like this, functional groups and specific local substructures can be identified. The featurization method results in a bit vector of user-defined length. Here, a length of 1024 is used with a radius set to 2, denoting the number of bonds considered in the neighborhood of an atom. With this radius, Morgan fingerprints are similar to the ECFP4 (Extended-Connectivity Fingerprints) representation, where a diameter of 4 is considered, while the algorithm for creation differs slightly.

Third and last, **CheMeleon embeddings** are used as a distinct set of features. These embeddings are retrieved from the CheMeleon foundation model, which is

⁴<https://github.com/scikit-learn/scikit-learn>

⁵<https://github.com/rdkit/rdkit>

pre-trained on a large unlabeled corpus from PubChem, featurized into Mordred descriptors [43]. The model leverages a D-MPNN (Direct Message-Passing Neural Network) to predict these descriptors. CheMeleon embeddings are available with a fixed length of 2048. This exceeds the maximum allowed feature size of TabPFN, which is why the TabPFN model is not applied to CheMeleon embeddings, but only to RDKit descriptors and Morgan fingerprints in the experiments.

After featurization, the datasets were preprocessed further if needed, which was only necessary in the case of the NN model. Contrary to CatBoost and TabPFN, where feature normalization is handled internally, features needed to be standardized here. The standard score z of a sample x is calculated independently for each feature as

$$z = \frac{x - \mu}{s} \quad (3)$$

where μ is the mean and s is the standard deviation of the feature across all training samples. Therefore, a StandardScaler object from the scikit-learn⁶ library is fitted on the training data. Later, the same scaler is used to transform the test data. Only numerical features were scaled, whereas other features, especially binary features, were preserved.

Hyperparameter tuning

Hyperparameters were tuned on the training set using 3-fold cross validation on the Butina splits described above. For the NN, 60 iterations of random search were performed to find the best hyperparameter values for learning rate, weight decay, size of the hidden layer and dropout rate with a search space following Friesacher et al. [33] (see Table 4).

Hyperparameter	Search space
Learning rate	[1e-5, 0.1]
Hidden size	[5, 130]
Weight decay	[1e-5, 0.1]
Dropout rate	[0.3, 0.9]

Table 4: Search space for hyperparameter tuning of NN

Learning rate and weight decay were sampled from a logarithmic distribution between the lower and higher bound.

⁶<https://github.com/scikit-learn/scikit-learn>

For CatBoost, hyperparameters were tuned in a two-step process. First, early stopping was performed to find the optimal number of iterations, which represents the number of trees in the final model, with a maximum of 1000. As a result, obvious under- or overfitting can be detected. Second, other hyperparameters were determined using random search, including tree depth, learning rate, bootstrap type, random strength, grow policy and auto class weights, with the determined number of iterations from the previous step. The corresponding search space is shown in Table 5.

Hyperparameter	Search space
Depth	[4, 8]
Learning rate	[1e-6, 0.5]
Bootstrap type	Bayesian, Bernoulli, MVS
Random strength	[0, 2]
Grow policy	SymmetricTree, Depthwise, Lossguide
Auto class weights	None, Balanced

Table 5: Search space for hyperparameter tuning of CatBoost

The learning rate was sampled from a logarithmic distribution between the lower and higher bound.

In the end, model performance of CatBoost with tuned hyperparameters was compared to its counterpart with standard hyperparameters in terms of accuracy and uncertainty metrics. The version with standard parameters outperformed the tuned version predominantly, so that tuning of CatBoost was omitted for this study.

For TabPFN, no tuning is required and standard model hyperparameters were used.

Model training

After setting the hyperparameters, each model was trained on the whole training set. The NN was trained for 50 epochs on the full training set batch. Binary cross-entropy was set as the loss function. An Adam optimizer was used with the tuned learning rate differing by dataset.

CatBoost and TabPFN were trained with standard parameters. Notably, fitting a TabPFN classifier does not actually train the model in terms of weight updates, but prepares internal processing of the train dataset for conditioning the fully pre-trained model on unseen data.

Ensemble models are trained on the same (tuned) hyperparameters as the single models.

Post-hoc calibration

As this study aims to compare the quality of uncertainty estimates across different models and data representations, it is crucial that predicted probabilities are well calibrated, ensuring that uncertainty measures are meaningful and comparable. A model with good calibration outputs probabilities that match real-world likelihoods. E.g., if the model predicts a 70% chance of an event to happen, in reality, the event should indeed occur 70% of the time. Recent work has shown that ML models are typically miscalibrated [44]. To increase calibration and therefore, the reliability of ML models, post-hoc calibration techniques have been developed. In this study, it is investigated how temperature scaling (TS) [45] improves calibration in the given scenario.

TS introduces a single parameter, the so-called temperature $T > 0$, that rescales the models logits z_i before applying softmax to obtain the probability p_i of class i :

$$p_i = \text{softmax}\left(\frac{z_i}{T}\right) \quad (4)$$

without changing the predicted class label and thus, the accuracy, of the model. The temperature value is optimized on a calibration set, minimizing the negative log-likelihood. Therefore, TS is a variant of Platt scaling [46]. In this study, TS was applied by using the probmetrics package⁷ [47].

Metrics

To comprehensively assess predictive performance and reliability of predictions, a set of established performance and UQ metrics was calculated. First of all, balanced accuracy was used to quantify predictive performance of the classification models. It accounts for any class imbalances present, where raw accuracy can be misleading. Balanced accuracy is defined as the average of recall obtained on each class, with a range of $[0, 1]$. The higher the balanced accuracy score, the better is the predictive performance of the classifier.

Next, it was evaluated how well the confidence score, which equals the maximum class probability predicted per sample, reflects model uncertainty. Therefore, the

⁷<https://github.com/dholzmueller/probmetrics>

Expected Calibration Error (ECE) [48] and Brier score [49] were calculated before and after applying TS. ECE is a scalar summary statistic of calibration. It measures miscalibration as the difference in expectation between confidence and accuracy. Predictions are either partitioned into equally-spaced or equal-mass bins. Here, M equally-spaced bins B_m are used. Per bin, the difference between confidence and accuracy is calculated and weighted by the number of samples in this bin:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (5)$$

where n is the total number of samples across all bins, with $\text{acc}(B_m)$ being defined as follows:

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbf{1}(\hat{y}_i = y_i) \quad (6)$$

with $\hat{y}_i$ as the predicted and y_i as the true class label for sample i . $\text{Conf}(B_m)$ is defined as

$$\text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i \quad (7)$$

with confidence $\hat{p}_i$ for sample i [45]. Here, ECE was computed using the class-1 probabilities with 10 bins ($M = 10$).

The Brier score measures the mean squared difference between predicted probabilities p_i for the positive class and the true outcomes y_i at sample i :

$$\text{BS} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2 \quad (8)$$

with n being the total number of predictions (samples) [50]. In that way, it evaluates both predictive performance and calibration.

Both Brier score and ECE have a range of $[0, 1]$. The lower the values of ECE and Brier score, the better is the calibration of the respective model and therefore, the more applicable is the confidence score as a measure of uncertainty.

Additionally, calibration curves were constructed by partitioning predicted class-1 probabilities into 10 equally spaced bins (analogous to the ECE computation) and comparing the average predicted confidence with the observed empirical accuracy within each bin.

Another UQ metric that was calculated is predictive entropy, which is the Shannon entropy H [51] computed on the model’s probabilistic output distribution over classes $i = 1, \dots, c$:

$$H = - \sum_{i=1}^c p_i \log p_i \quad (9)$$

The entropy value is calculated per sample. It will be aggregated here to the mean across all samples i in the test set T to obtain a scalar value:

$$\text{Mean Predictive Entropy} = \frac{1}{|T|} \sum_{i \in T} H_i \quad (10)$$

For the ensemble models, various ensemble-specific metrics can be computed, including the voting ratio, the variance of predicted probabilities or the ensemble disagreement. In this study, the ensemble disagreement score via the Kullback-Leibler divergence, as suggested by Lakshminarayanan et al. [52], was used to evaluate the uncertainty of model types via ensembling. The disagreement score is defined as follows:

$$\text{Disagreement} = \sum_{m=1}^M \text{KL}(p_{\theta_m}(y | x) || p_E(y | x)) \quad (11)$$

where KL denotes the Kullback-Leibler divergence [53], p_{θ_m} is the probabilistic predictive distribution of a member m over classes and p_E is the average of predicted probabilities, which represents the prediction of the ensemble. This score likewise yields per-sample values and can be aggregated to the mean disagreement across samples.

Uncertainty quantification experiments

Given the described datasets, representations and models, different experiments were performed to evaluate performance and UQ quality across scenarios, that is, each combination of the above (see Figure 5). As an example, the CA-IV dataset was transformed to RDKit descriptors in order to train CatBoost. Due to the exclusion of CheMeleon embeddings with TabPFN for all three datasets, this results in 24 different scenarios.

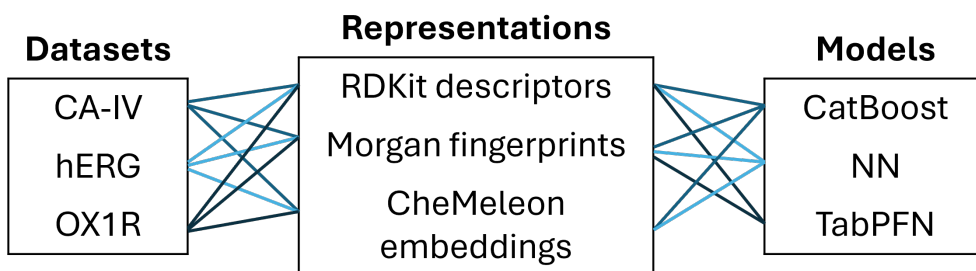


Figure 5: Experiment scenarios for uncertainty quantification

Excluding the combination of CheMeleon embeddings and TabPFN, every combination of dataset, representation and model is considered a scenario.

Per scenario, 10 repeated training runs were performed with a different seed each and a bootstrapped subset of the training data made available to capture variability across random initialization in training as well as data variability. To obtain the bootstrapped subsets of the training data, the original train set was resampled to the same length using sampling with replacement. For the ensemble models, only one training run was performed per model type to reduce computational cost. Ensemble runs already cover the training of multiple models. The aggregation of outcomes across ensemble members inherently decreases variance and stabilizes performance and uncertainty estimates.

Statistical testing

The statistical significance of differences for each evaluation metric was assessed using a Friedman-Nemenyi testing procedure. First, the Friedman test was applied to show whether any significant differences exist between the three models in each experiment scenario. The test ranks the respective metric per model and evaluates whether average ranks differ more than expected by chance.

If the Friedman test indicated a significant overall difference, a Nemenyi post-hoc test was conducted to perform pairwise comparisons between models and identify those pairs with significant differences in the respective metric. The significance threshold for p-values was set to $\alpha = 0.05$. Statistical significance levels are reported using star annotations according to the corresponding p-value thresholds: * ($0.01 < p \leq 0.05$), ** ($0.001 < p \leq 0.01$), *** ($0.0001 < p \leq 0.001$), and **** ($p \leq 0.0001$).

The tests were performed using the Pingouin⁸ package and the scikit-posthocs⁹ package.

⁸<https://pingouin-stats.org/index.html>

⁹<https://scikit-posthocs.readthedocs.io/en/latest/#>

3.3 Active learning as a downstream task

Active learning (AL) is performed as a downstream task following UQ evaluation. When applying uncertainty-based acquisition, the AL effectiveness fundamentally depends on the quality of the uncertainty estimates produced by the model that is used. The purpose of UQ is to quantify the model’s confidence in its predictions, and thereby to identify regions of the input space where the model lacks sufficient knowledge. This corresponds to high epistemic uncertainty arising from limited or missing training data. AL leverages these uncertainty estimates to guide the data acquisition process by preferentially selecting the most informative unlabeled samples. Therefore, UQ provides the decision criterion, while AL defines the action that is based on this criterion.

Evaluating AL performance serves as a practical and task-oriented way to assess the usefulness of uncertainty estimates. Well-calibrated and informative uncertainties should enable more efficient sample selection and faster improvements in model performance compared to random sampling.

Data regime

After conducting the initial experiments across multiple datasets, the subsequent AL analyses were continued using only the largest of the three datasets, which is the OX1R dataset in particular. This decision was made because the larger dataset provides a more reliable and representative basis for evaluating different query methods, as the greater number of samples ensures a sufficiently large unlabeled pool.

The time-based train-test split provided in the dataset was reused to separate the held-out test set across all AL iterations. The initially labeled pool, referred to as the starting budget, was then split randomly from the train data. No extra calibration set was used, as post-hoc calibration was not applied in this part of the experiment.

As a feature representation, RDKit descriptors were chosen for all AL experiments, as they performed better than Morgan fingerprints in three out of four metrics for OX1R. This was tested statistically using a Wilcoxon signed-rank test across 30 paired runs (see Table 6). CheMeleon embeddings were excluded from the comparison due to their incompatibility with TabPFN.

Metric	Hypothesis	W statistic	p-value	Significance
Balanced accuracy ($\uparrow$)	RDKit > Morgan	401	3.61×10^{-5}	Significant
ECE ($\downarrow$)	RDKit < Morgan	246	0.612	Not Significant
Brier score ($\downarrow$)	RDKit < Morgan	49	2.48×10^{-5}	Significant
Mean predictive entropy ($\downarrow$)	RDKit < Morgan	126	0.0139	Significant

Table 6: Wilcoxon signed-rank test results on OX1R comparing RDKit descriptors and Morgan fingerprints

The Wilcoxon signed-rank test was performed on 30 paired runs (3 models x 10 repeats). For Wilcoxon signed-rank test, the W statistic is the sum of the ranks of signed differences between paired observations and indicates whether the differences consistently favor one method over the other.

Lüth et al. [54] define different label regimes concerning the starting budget and query size in AL: a low-label, mid-label and high-label regime. In this study, the mid-label regime was chosen to allow for meaningful performance increases per iteration, while avoiding cold start problems and limiting the number of iterations until the unlabeled pool is exhausted. Particularly, starting budget and query size are set to $25 \times C$, where C denotes the number of classes. As binary classification with two classes is performed here, this results in 50 samples.

Models

Following the UQ experiments, the models evaluated for AL were CatBoost, TabPFN and the baseline NN. CatBoost and TabPFN were still used with default hyperparameters, whereas the hyperparameters of the NN (see Table 4) were tuned per iteration on the respective labeled pool.

In addition to these models, several hybrid models were constructed from the single models and evaluated to enable a more thorough comparison of query methods. Each hybrid model consists of an embedding extractor trained on the input features to obtain latent feature representations, and a prediction head taking this latent representation as an input for the classification task. There are two types of embeddings extractors implemented here: one for TabPFN embeddings and one for CatBoost embeddings.

TabPFN embeddings can be directly extracted from a trained TabPFN model

after fitting it once on the entire train dataset. This yields a list of embeddings from each of the TabPFN estimators. In this case, 8 estimators are used as defined in the default hyperparameters and their embeddings are concatenated for each sample.

CatBoost does not have an inbuilt embedding extractor as it is a GBDT model. Instead, leaf indices are calculated, representing which leaf a sample ends up in for each tree of the tree model. In the next step, leaf indices are one-hot encoded to obtain a binary embedding vector of fixed length.

On top of both of the embedding extraction modules, a NN of the same architecture as used before was trained as a prediction head to output the final predictions. Here, no hyperparameter tuning of the NN is implemented, so a fixed learning rate of 0.001, hidden size of 128, weight decay of 0 and dropout rate of 0.5 was set across the subsequent experiments. Additionally, CatBoost was used as prediction head for TabPFN embeddings with default hyperparameters. An overview of the three resulting models can be seen in Table 7.

Abbr.	Embedding extractor	Prediction head
embedding_cb	TabPFN	CatBoost
embedding_nn	TabPFN	NN
cb_nn	CatBoost	NN

Table 7: Hybrid models for active learning

Each hybrid model consists of an embedding extractor and a prediction head.

Query methods

Query methods define the strategy by which the most informative samples from the unlabeled pool are selected for annotation in an AL iteration. The comparison of different query methods is central to this study. As query methods can vary substantially in their ability to select the most informative samples, comparing these methods allows assessing how effectively uncertainty estimation can be leveraged to improve learning.

The baseline that every query method was compared against is the **random** acquisition strategy. It draws samples from the pool in a completely random manner. The first uncertainty-based query method investigated in this study was

the **entropy** strategy. It selects the samples with the highest predictive entropy as defined in Equation (9).

Next, **BALD** (Bayesian Active Learning by Disagreement) [55] defines a query method that makes use of mutual information between the predicted class labels and the model parameters. It relies on a posterior distribution of class predictions, that can be introduced by Monte Carlo (MC) dropout. An alternative for non-neural models is to produce the posterior distribution with an ensemble, e.g. by using truncated model versions of the trained CatBoost model as a so-called virtual ensemble here. A related form of BALD is **Bregman Information (BI)** [56]. BI measures prediction variability in a dataset $Z = [z_1, \dots, z_n]$ and can be estimated by

$$\text{BI}_{\text{LSE}} = \frac{1}{n} \sum_{i=1}^n \text{LSE}(z_i) - \text{LSE}\left(\frac{1}{n} \sum_{i=1}^n z_i\right) \quad (12)$$

based on BI of a random variable Z generated by $G = \text{LSE}$ (LogSumExp):

$$\text{BI}_G = \mathbb{E}[G(Z)] - G(\mathbb{E}[Z]) \quad (13)$$

BI is derived from Bregman divergence as a generalized notion of variance induced by a convex function. For both BI and BALD acquisition, different values of n were evaluated, representing the number of samples drawn to span the posterior distribution. The results are depicted in Appendix A.

Another query method that is commonly used and was applied in this study is **BADGE** (Batch Active Learning by Diverse Gradient Embeddings) [57]. It considers both predictive uncertainty and sample diversity. In this regard, it first computes gradient embeddings with respect to the last layer parameters and then selects points by sampling from the gradient embedding space by incorporating the K-means++ initialization scheme. BADGE is only applicable to NN models or hybrid models with a NN head.

Additionally, for CatBoost models, **CatBoost uncertainty** is used as the basis for a query method. It can be derived directly from a CatBoost model, using a virtual ensemble prediction [58]. From the total uncertainty and aleatoric uncertainty (data uncertainty) score the model returns per sample, one can infer the epistemic uncertainty (knowledge uncertainty), which is of special interest for AL. The query method then chooses samples with highest epistemic uncertainty to be labeled.

Lastly, a new query method was implemented to leverage uncertainty estimates by incorporating **diversity** of pool samples measured in the input descriptor space. Therefore, samples are clustered in the feature space using HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) [59] with a minimum cluster size of 10. The parameter was selected to avoid acquisitions driven by singleton or near-singleton clusters, which would correspond to outliers rather than informative unexplored regions. In each cluster, samples are ranked by highest predictive entropy. In a round-robin fashion, samples are then selected from each cluster until the query size is reached, acquiring the most uncertain point in each cluster per round (see algorithm 1). Noise data points are treated as one separate cluster with label "-1".

Algorithm 1 Diversity-Based Acquisition via HDBSCAN Clustering

Require: Pool set feature matrix X_{pool} , predictive probabilities $\mathbf{p}$, query size q
Ensure: Set of selected indices $\mathcal{A}$

```

 $\mathbf{c} \leftarrow \text{HDBSCAN}(X_{\text{pool}})$  ▷ Cluster labels
 $\mathbf{u} \leftarrow \text{PredictiveEntropy}(\mathbf{p})$  ▷ Uncertainty score per sample
 $\mathcal{C} \leftarrow \{\}$ 
for  $k \in \mathbf{c}$  do
     $I_k \leftarrow \{i : c_i = k\}$  ▷ Indices belonging to cluster  $k$ 
     $\mathcal{C}[k] \leftarrow \text{sort}(I_k, \text{key} = -\mathbf{u})$  ▷ Sort descending by uncertainty
end for
 $\mathcal{A} \leftarrow []$  ▷ Selected indices
while  $|\mathcal{A}| < q$  do
    for  $k \in \mathcal{C}$  do
        if  $\mathcal{C}[k] \neq \emptyset$  then
             $i^* \leftarrow \text{pop}(\mathcal{C}[k])$  ▷ Take most uncertain sample
             $\mathcal{A} \leftarrow \mathcal{A} \cup \{i^*\}$ 
            if  $|\mathcal{A}| = q$  then
                break
            end if
        end if
    end for
end while
return  $\mathcal{A}$ 

```

Active learning experiments

To evaluate the query methods described above, several experiments were conducted. Per (hybrid) model and each query method applicable for that model, 10 repeated runs with different random starting budgets of the same size were

performed per run. In each run, AL cycles were iterated until the pool was exhausted, which results in 11 iterations in total. As the pool size is not divisible by the query size, there is a lower budget increase in the last iteration, where all remaining samples are acquired.

Evaluation

For evaluation, learning curves over AL cycles were plotted, showing the iteration-wise progression of balanced accuracy. The balanced accuracy of the last AL cycle equals the full dataset performance of the model. Furthermore, the area under the learning curve (AULC) was computed by using trapezoidal integration of the learning curve over the labeling budget. This enables a comparison of methods by a point value.

Differences in AULC values were then compared statistically by applying Friedman’s tests and post-hoc Nemenyi tests to obtain p-values per pair of query methods within each model.

3.4 Computational environment

The experiments for this study were conducted on a shared compute cluster node equipped with 80 Intel Xeon Gold CPU cores. For implementation, Python 3.13 was used. Shell scripts were executed via GNU bash (version 5.2). All program code was executed inside a virtual environment managed with Pip and Conda to ensure reproducibility. The respective requirements file and the complete program code, including experiment configurations, can be accessed through the GitHub repository¹⁰.

¹⁰<https://github.com/MLO-lab/uncertainty-chem>

4 Results

This study aims to assess the reliability of molecular predictions. To this end, several uncertainty quantification metrics were applied and evaluated on multiple molecular datasets and ML models. In addition, the potential of uncertainty estimates to guide active learning strategies for improved data acquisition was investigated. The results of these analyses are presented in the following.

4.1 Predictive performance of classification models

First, the predictive performance of CatBoost, the baseline NN and TabPFN was assessed for every experiment scenario using balanced accuracy as performance metric. Although model performance does not evaluate reliability of predictions, it is essential to assess the usefulness of a model. As shown in Figure 6, all models perform with accuracies between 60 and 90% across datasets and representations. Overall, there are barely significant differences between model types per scenario.

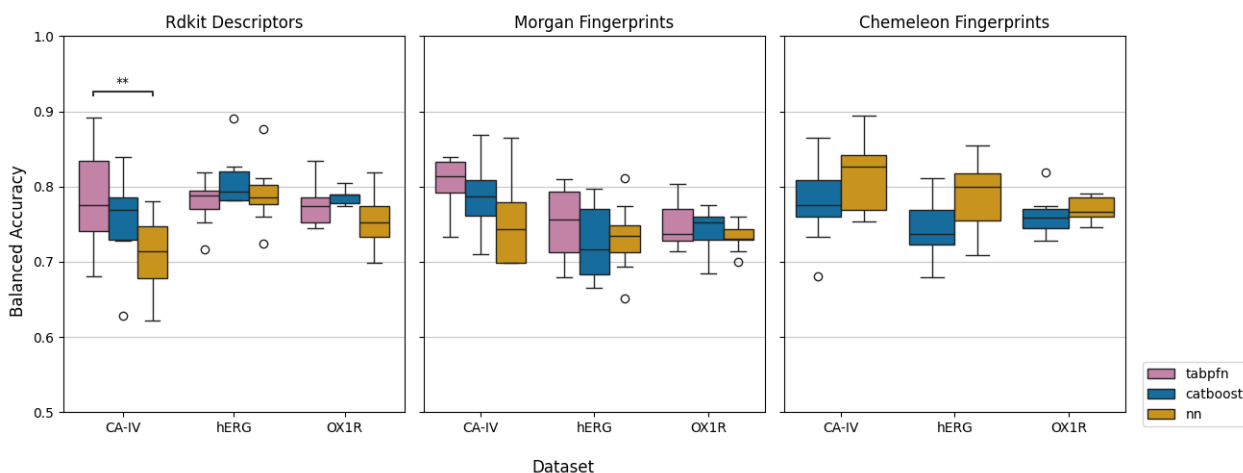


Figure 6: Balanced accuracy (↑) comparison

The distribution of balanced accuracy values across repeated runs are shown per experiment scenario. Brackets indicate statistical significance between models. Significance levels follow the notation defined in section 3.2.

The ensemble counterparts of the models perform comparably to the base versions, with balanced accuracies shown in Table 8.

Dataset	Representation	CatBoost	NN	TabPFN
CA-IV	RDKit Descriptors	0.76	0.73	0.81
CA-IV	Morgan Fingerprints	0.76	0.70	0.87
CA-IV	CheMeleon Fingerprints	0.71	0.83	-
hERG	RDKit Descriptors	0.80	0.83	0.81
hERG	Morgan Fingerprints	0.77	0.74	0.78
hERG	CheMeleon Fingerprints	0.79	0.82	-
OX1R	RDKit Descriptors	0.77	0.76	0.79
OX1R	Morgan Fingerprints	0.75	0.73	0.76
OX1R	CheMeleon Fingerprints	0.79	0.80	-

Table 8: Balanced accuracy ($\uparrow$) of ensemble models

Per model type, an ensemble of 5 members was evaluated in a single repeat. Per scenario, the highest balanced accuracy value is denoted in bold, highlighting the best performing ensemble.

4.2 Calibration and reliability of predictive probabilities

Model calibration was evaluated to assess the reliability of predictive probabilities as uncertainty estimates using different techniques.

Visualization of model calibration with calibration curves

First, calibration curves were visualized to show the relationship between predicted confidence and accuracy. As shown in Figure 7, the curves mostly lie below the diagonal, indicating that all models are overconfident in their predictions to some extent. The highest deviation from the diagonal, marking the worst calibration, can be seen for the smallest dataset CA-IV (first row) and the CheMeleon embedding representation (last column).

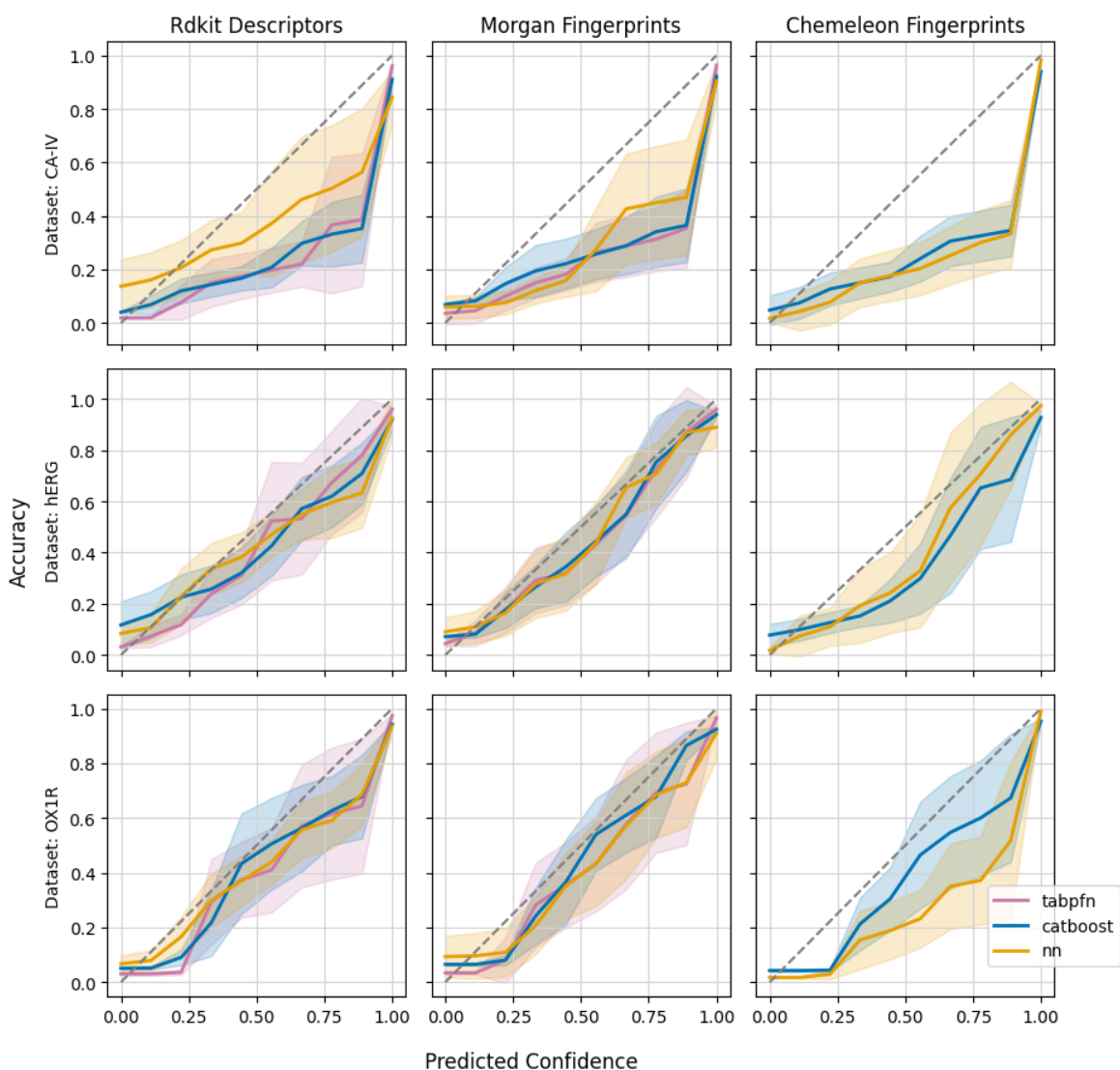


Figure 7: Calibration curves

Reliability of predicted confidence is presented in calibration curves across all datasets (rows) and representations (columns). The diagonal line references perfect model calibration. For each model, mean curves with standard deviation across 10 repeated runs are shown.

Evaluation of calibration using ECE and Brier score

To be able to compare calibration of models per scenario more precisely, ECE and Brier score were calculated. Results for the single models are presented in Figure 8 and 9, respectively.

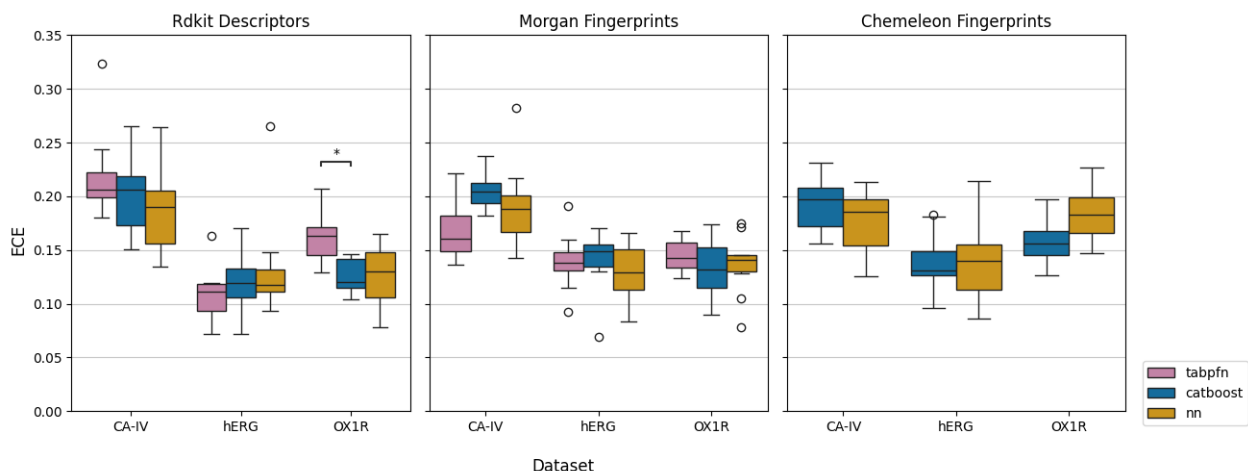


Figure 8: ECE (↓) comparison

The distribution of ECE values across repeated runs are shown per experiment scenario. The closer the value is to 0, the better the model is calibrated. Brackets indicate statistical significance between models. Significance levels follow the notation defined in section 3.2.

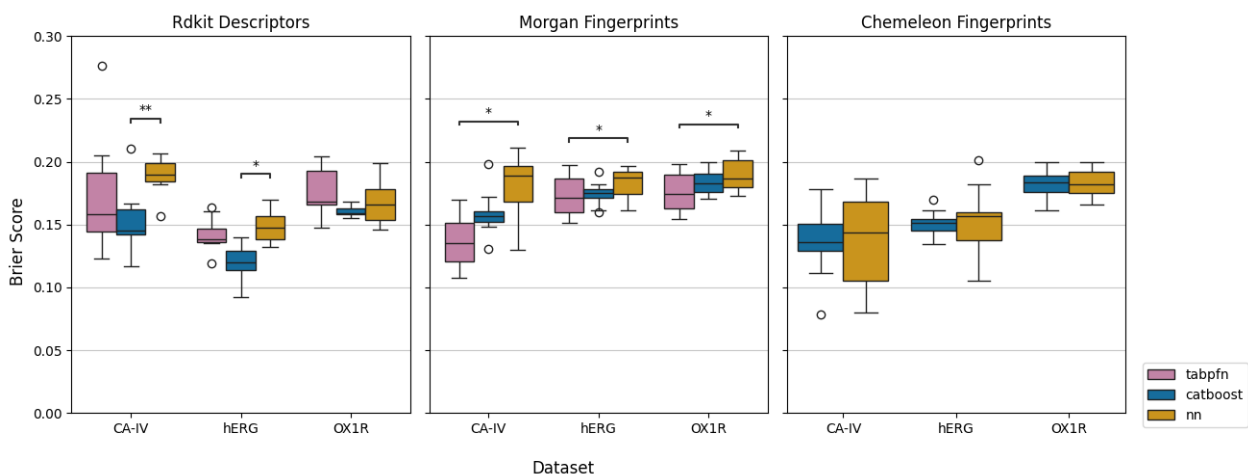


Figure 9: Brier score (↓) comparison

The distribution of Brier scores across repeated runs are shown per experiment scenario. The closer the value is to 0, the better the model is calibrated. Brackets indicate statistical significance between models. Significance levels follow the notation defined in section 3.2.

The comparison of ECE values confirms the results of the calibration curves (see Figure 7), showing lower values for the larger datasets, hERG and OX1R. Between models, there are barely significant differences. Brier scores, however, show slightly different behavior, with barely any improvements for hERG and OX1R compared to the CA-IV dataset.

The resulting ECE values for the ensemble runs are shown in Table 9. Corresponding Brier scores are listed in Table 10.

Dataset	Representation	CatBoost	NN	TabPFN
CA-IV	RDKit Descriptors	0.172	0.234	0.163
CA-IV	Morgan Fingerprints	0.173	0.155	0.147
CA-IV	CheMeleon Fingerprints	0.173	0.128	-
hERG	RDKit Descriptors	0.147	0.155	0.088
hERG	Morgan Fingerprints	0.122	0.172	0.104
hERG	CheMeleon Fingerprints	0.120	0.122	-
OX1R	RDKit Descriptors	0.129	0.078	0.141
OX1R	Morgan Fingerprints	0.162	0.131	0.133
OX1R	CheMeleon Fingerprints	0.171	0.152	-

Table 9: ECE ($\downarrow$) of ensemble models

Per model type, an ensemble of 5 members was evaluated in a single repeat. Per scenario, the lowest ECE value is denoted in bold, highlighting the best performing ensemble regarding calibration error.

Dataset	Representation	CatBoost	NN	TabPFN
CA-IV	RDKit Descriptors	0.137	0.181	0.131
CA-IV	Morgan Fingerprints	0.148	0.181	0.120
CA-IV	CheMeleon Fingerprints	0.124	0.095	-
hERG	RDKit Descriptors	0.117	0.135	0.131
hERG	Morgan Fingerprints	0.160	0.174	0.152
hERG	CheMeleon Fingerprints	0.135	0.120	-
OX1R	RDKit Descriptors	0.150	0.160	0.155
OX1R	Morgan Fingerprints	0.175	0.186	0.164
OX1R	CheMeleon Fingerprints	0.169	0.158	-

Table 10: Brier score ($\downarrow$) of ensemble models

Per model type, an ensemble of 5 members was evaluated in a single repeat. Per scenario, the lowest Brier score is denoted in bold, highlighting the best performing ensemble.

Effect of post-hoc calibration on probability estimates

In the uncertainty pipeline, TS was applied as a post-hoc calibration technique using a hold-out calibration set to reduce calibration error. The effects are shown in Figure 10. For most scenarios and models, calibration cannot be improved substantially. One exception to highlight is the improvement of NN calibration when OX1R with the CheMeleon embedding representation is used.

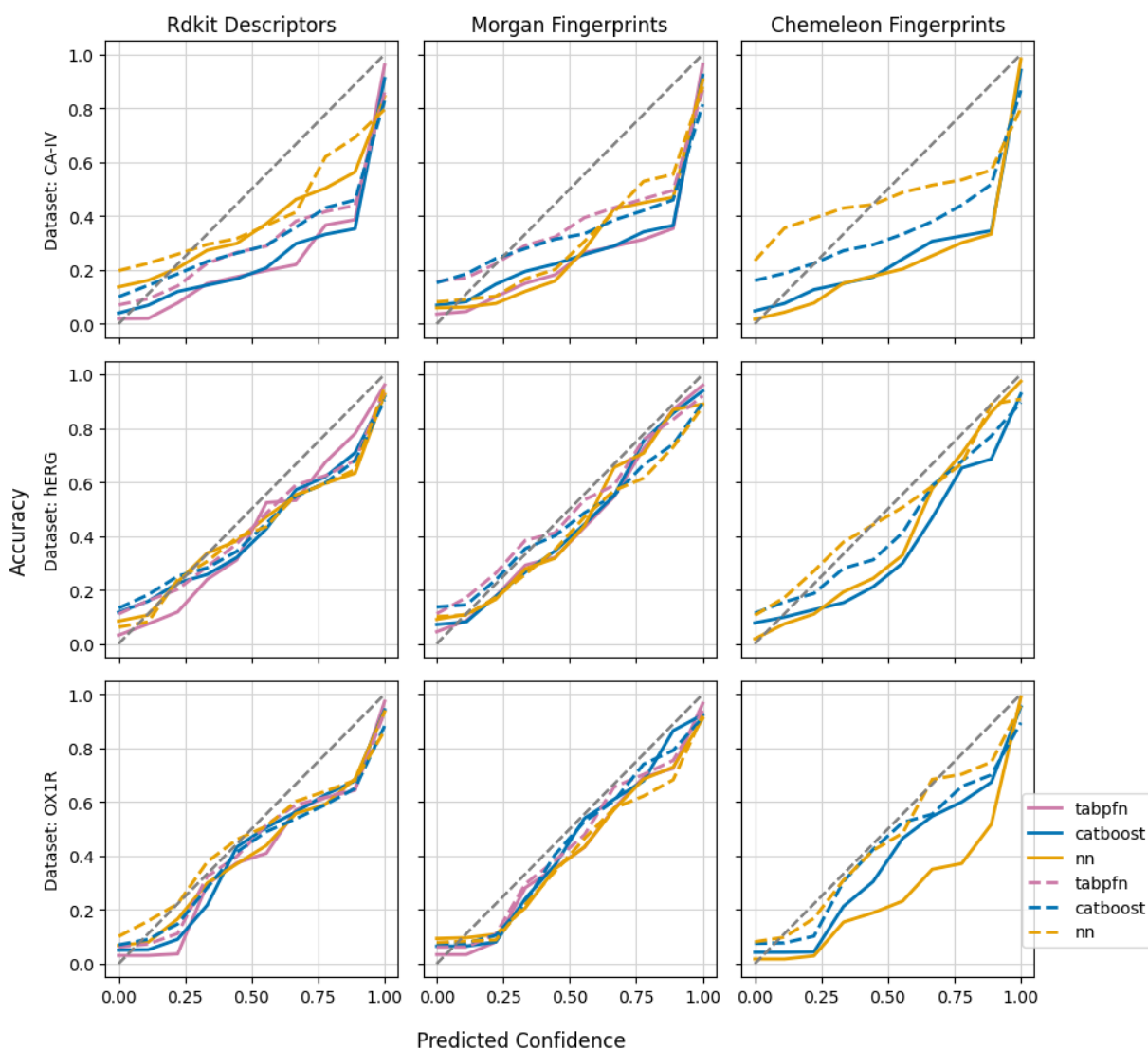


Figure 10: Effects of temperature scaling

Reliability of predicted confidence before (solid lines) and after the application of TS (dashed lines) is presented in mean calibration curves across all datasets (rows) and representations (columns). The diagonal line references perfect model calibration.

4.3 Characteristics of predictive uncertainty

Predictive uncertainty was measured using both mean predictive entropy and disagreement of ensemble members as defined in Equation (9) and (11). For mean predictive entropy, a repeating pattern can be observed (see Figure 11). Mean entropy values are lowest for TabPFN throughout all scenarios, indicating lower uncertainty. The NN is most uncertain in its predictions with values closer to 0.7, which would arise in case of a random guess. However, for CheMeleon embeddings, the pattern is reversed, as the NN outperforms CatBoost there in terms of mean predictive entropy.

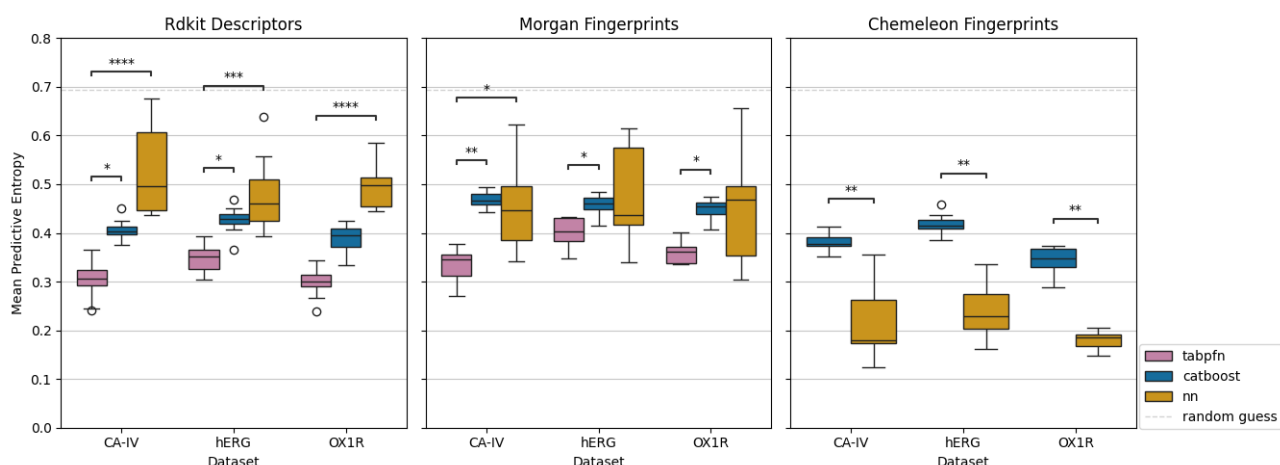


Figure 11: Mean predictive entropy comparison

Predictive entropy is evaluated and averaged across samples per run. The distribution across repeated runs are shown per experiment scenario. The closer the value is to 0, the more certain is the model in its predictions. Brackets indicate statistical significance between models. Significance levels follow the notation defined in section 3.2.

Mean predictive entropy values for the ensemble runs are shown in Table 11. Notably, the TabPFN ensemble also outperforms the other models regarding mean predictive entropy. For CheMeleon embeddings, where TabPFN cannot be applied, the NN has lower uncertainty than CatBoost, which is consistent with the observations for single models.

Dataset	Representation	CatBoost	NN	TabPFN
CA-IV	RDKit Descriptors	0.448	0.606	0.392
CA-IV	Morgan Fingerprints	0.492	0.508	0.400
CA-IV	CheMeleon Fingerprints	0.417	0.273	-
hERG	RDKit Descriptors	0.434	0.436	0.376
hERG	Morgan Fingerprints	0.477	0.451	0.420
hERG	CheMeleon Fingerprints	0.449	0.292	-
OX1R	RDKit Descriptors	0.399	0.521	0.353
OX1R	Morgan Fingerprints	0.455	0.404	0.362
OX1R	CheMeleon Fingerprints	0.357	0.209	-

Table 11: Mean predictive entropy ($\downarrow$) of ensemble models

Per model type, an ensemble of 5 members was evaluated in a single repeat. Per scenario, the lowest mean entropy score is denoted in bold, highlighting the most certain ensemble.

Whereas predictive entropy characterizes total predictive uncertainty, ensemble disagreement offers insight into uncertainty originating from differences between

individual ensemble members. Figure 12 therefore examines ensemble disagreement across experiment scenarios. The median values across models barely differ, whereas the distributions of values across samples differ significantly in some cases. Here, the NN seems to be more uncertain for CheMeleon embeddings, while CatBoost predicts most stable across ensemble members, indicating low uncertainty.

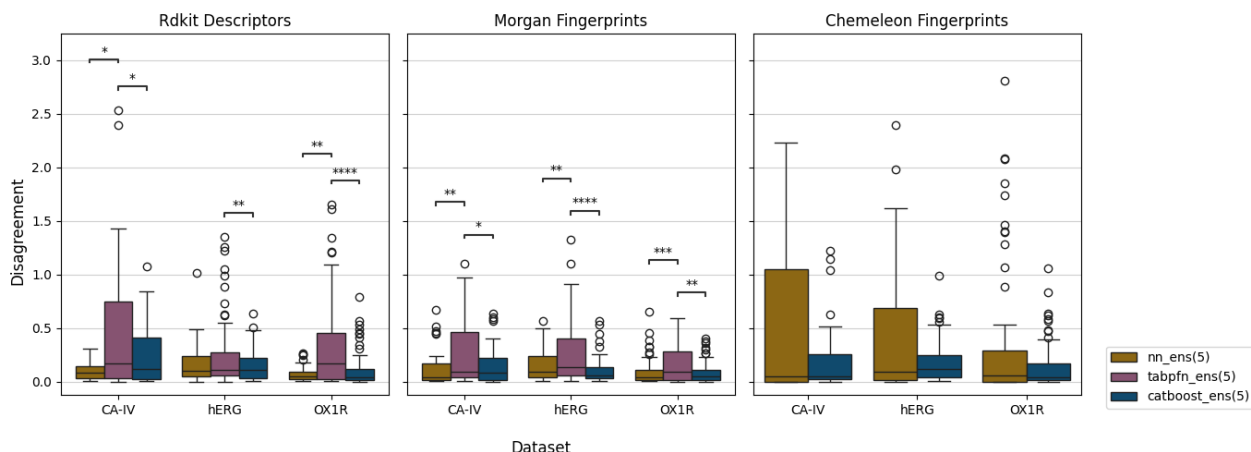


Figure 12: Per-sample distribution of ensemble disagreement

The distribution of ensemble disagreement scores over samples is shown for the ensemble run. A value of 0 would indicate perfect agreement on predictive class probabilities among ensemble members. Brackets indicate statistical significance between models. Significance levels follow the notation defined in section 3.2.

Before discussing what these results mean for the reliability of molecular predictions, the AL experiments will be evaluated.

4.4 Learning efficiency of query methods in active learning

The preceding results demonstrated differences in calibration quality and uncertainty characteristics across models. An important remaining question is whether high-quality uncertainty estimates lead to improved data efficiency in practical learning scenarios. To address this, uncertainty-based active learning strategies are evaluated in the following section.

First, learning curves are visualized for the three base models (Figure 13) and three hybrid models (Figure 14), showing how much balanced accuracy increases per AL iteration. In general, curves increase steadily for all models. Regarding the base models, especially TabPFN and CatBoost display typical improvements in the first iterations and plateau towards the last iterations,

indicating good label efficiency. For the NN, the plateau arises earlier, ending in lower overall performance.

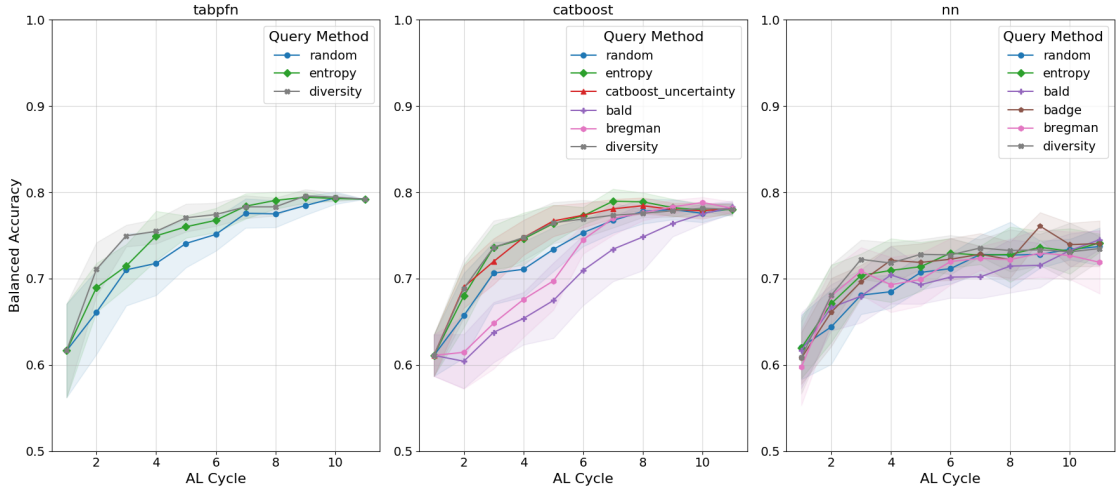


Figure 13: Learning curves (base models)

Mean learning curves are shown with standard deviation across 10 repeated runs with different random starting budgets for all base models. The random baseline method is depicted in blue.

The hybrid modeling approach, using an embedding extractor combined with a prediction head based on the base models, demonstrates efficacy, with maximum accuracies of over 80% (see Figure 14). Again, there are steady performance increases over iterations.

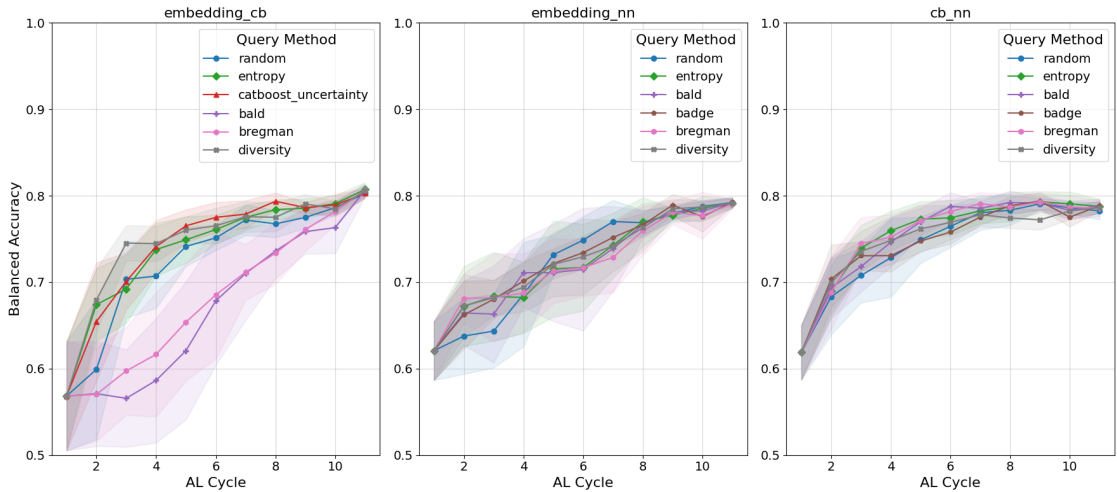


Figure 14: Learning curves (hybrid models)

Mean learning curves are shown with standard deviation across 10 repeated runs with different random starting budgets for all hybrid models. The random baseline method is depicted in blue.

To evaluate whether uncertainty-based query methods can outperform the

random baseline strategy, AULC values were calculated per model and query method. Results are shown in Figure 15 and 16 for the base models and hybrid models, respectively.

The AULC comparison confirms that random querying is a good baseline for AL. The diversity method, however, outperforms random with statistical significance in two of three base models, being TabPFN and the NN. Notably, general differences are minor, with a total range of 60% to 80% of AULC.

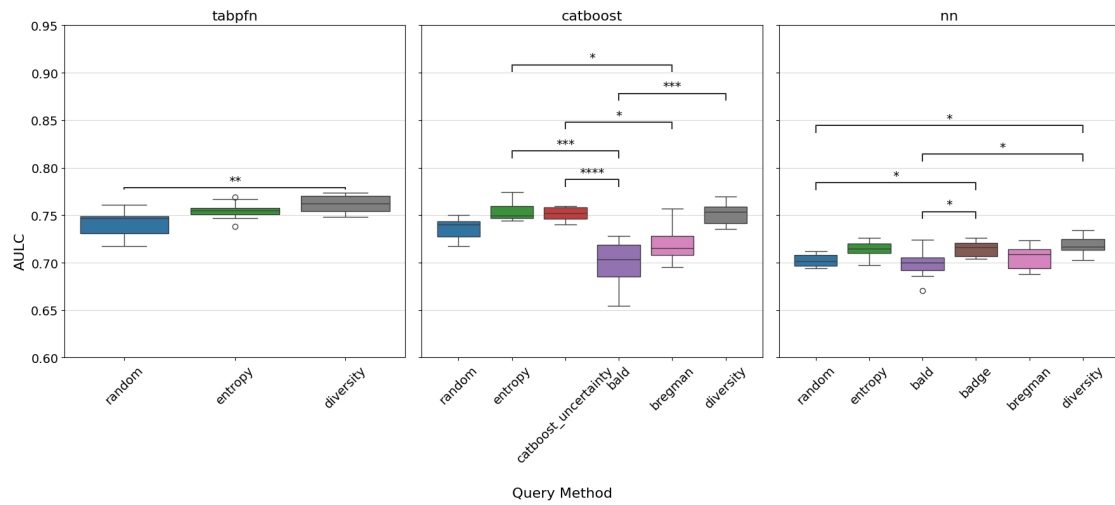


Figure 15: AULC comparison (base models)

The distribution of AULC values across 10 repeated runs with different random starting budgets are shown per applied query method for all base models. Brackets indicate statistical significance between models. Significance levels follow the notation defined in section 3.2.

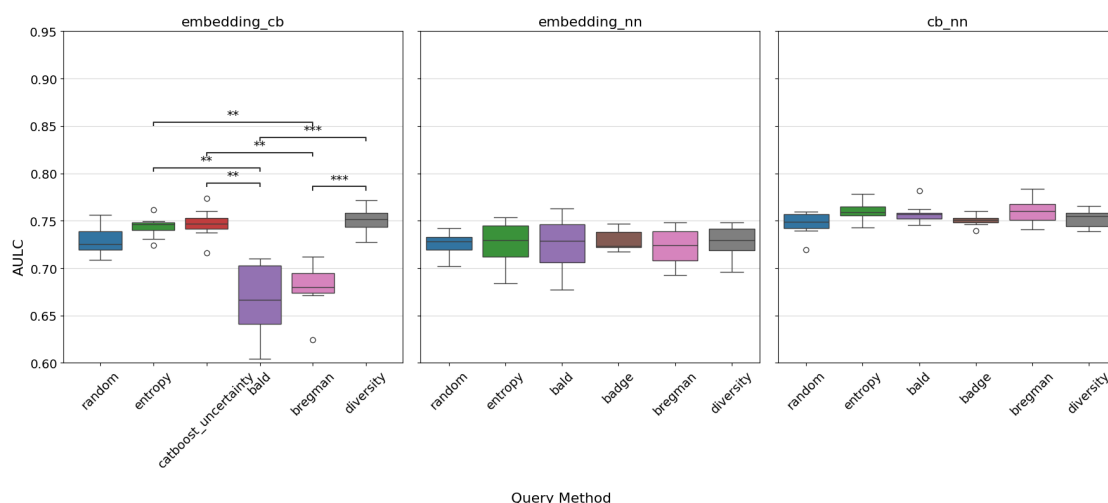


Figure 16: AULC comparison (hybrid models)

The distribution of AULC values across 10 repeated runs with different random starting budgets are shown per applied query method for all hybrid models. Brackets indicate statistical significance between models. Significance levels follow the notation defined in section 3.2.

These results provide the basis for interpreting the behavior of models and uncertainty estimates under different conditions, revealing reliability of molecular predictions, which will be discussed in the next section.

5 Discussion

This chapter discusses the main findings of this work, critically reflects on its limitations and practical implications, and outlines potential directions for future research.

5.1 Main findings

Overall, the results demonstrate that predictive performance is barely affected by model architecture and feature representation (Figure 6). However, calibration quality is influenced by both model type and representation (Figure 7), reflecting that predicted confidence scores are not equally reliable across scenarios. Looking at Brier score, especially the NN strikes with significantly higher values compared to CatBoost using RDKit descriptors and compared to TabPFN using Morgan fingerprints (Figure 9). It predicts with overconfidence, which is a commonly observed behaviour of NN architectures, consistent with previous findings [60]. Even a small network as used in this study seems to produce overconfident predictions, as cross-entropy optimization encourages increased logit magnitudes without accounting for parameter uncertainty.

Predictive entropy (Figure 11) and ensemble disagreement (Figure 12) show differing patterns. Regarding mean predictive entropy across samples per run, the NN again seems to be outperformed by the other models in terms of uncertainty for two out of three representations. On the contrary, for CheMeleon embeddings, the NN performs with lower median values of mean predictive entropy compared to all other scenarios and models. For RDKit descriptors and Morgan fingerprints, TabPFN has the lowest value ranges, indicating low uncertainty. Ensemble disagreement, on the other hand, shows that TabPFN cannot hold low uncertainty values when implemented in an ensemble of multiple members. Here, the CatBoost ensemble is the most reliable predictor across scenarios according to disagreement. A reason for the differing patterns between mean predictive entropy and ensemble disagreement could be that entropy covers total uncertainty, while ensemble disagreement primarily captures the reducible epistemic uncertainty, as variability between independently trained models indicates uncertainty about the model parameters rather than intrinsic noise in the data [8].

Regarding the AL experiments, this study records that uncertainty measures can leverage AL towards better learning efficiency (see section 4.4), as the entropy

and CatBoost uncertainty methods both expose higher AULC values in most of the examined base and hybrid models than with random acquisition. However, the combination of uncertainty with diversity-based querying in the introduced diversity method performs significantly better than the random method in TabPFN and the NN, highlighting that uncertainty alone might not be sufficient when deciding which samples to label in AL.

5.2 Limitations and practical implications

Before processing, the datasets were analyzed towards chemical coverage using Bemis-Murcko scaffolds. While CA-IV and hERG expose no dominant structural motifs, in OX1R, similar scaffolds were observed, indicating structural redundancy (Figure 3). The dominance of specific scaffolds likely reflects the medicinal chemistry focus of the underlying laboratory experiments [26]. However, it is important to note that this could have promoted overfitting in the ML experiments with this dataset, at the risk of low generalization ability towards compounds with underrepresented structures.

For the featurization of the data, Morgan fingerprints were used with a length of 1024 and a fixed radius of 2, encoding the atomic environment of molecular compounds considering 2 bonds in the neighborhood of each atom in 1024 bits. That means there was no comparison to Morgan fingerprints usage with different parameters, which might influence reliability and active learning efficiency as well as other criteria such as model architecture and dataset constitution. Another extension left for future experiments is the combination of different molecular representations, e.g. by concatenating RDKit descriptors and Morgan fingerprints.

Concerning the assessment of calibration quality of different models for the described datasets and data representations, ECE (Figure 8) and Brier score (Figure 9) were calculated. ECE provides an intuitive measure of calibration by directly comparing confidence and empirical accuracy, but suffers from binning sensitivity and lacks proper scoring properties. In contrast, Brier score constitutes a proper scoring rule that evaluates overall probabilistic prediction quality, combining calibration and discrimination effects, which may lead to discrepancies between both metrics. Brier Score therefore does not purely measure calibration, as well-calibrated models according to ECE might still lack discriminative power.

To access calibration quality using Brier score more precisely, it can be decomposed into calibration loss and refinement loss [61]. An alternative approach

suggests decomposition into reliability, resolution, and uncertainty components [62]. Reliability quantifies calibration error, resolution measures the model’s ability to separate instances with different outcome frequencies, and uncertainty reflects intrinsic dataset difficulty. These decomposition approaches explain why improvements in Brier score may arise from enhanced discrimination rather than improved calibration alone. To increase calibration quality and therefore the reliability of predicted confidence, post-hoc calibration was performed using a held-out calibration set in the uncertainty study (Figure 10). Due to the limited dataset size, this calibration set was split from the predefined test set, which does not represent the train set distribution. This may have limited the effectiveness of TS here. In addition, the chosen method for post-hoc calibration, being TS, was developed for NNs on image data, which could explain why the effectiveness for tabular classifiers like CatBoost and TabPFN is reduced [63]. Subsequently, post-hoc calibration was omitted for the AL study, where the full test set was used to determine balanced accuracy.

In AL, this study aimed to leverage learning efficiency by uncertainty-based querying. In comparison to the random query method, only two methods were able to outperform this random baseline significantly across 10 repeated runs for all base and hybrid models: the diversity method for TabPFN and the NN as well as BADGE for the NN only (Figure 15). For the hybrid models, no uncertainty-based query method achieved statistically better performance than random sampling (Figure 16). As random sampling selects instances without attention to utility, but implicitly selects representative examples that are independent and identically distributed (i.i.d.), this is making it a strong baseline [64]. It has also been demonstrated that uncertainty-based AL can perform worse than random sampling when the machine learning model has low capacity and cannot cover the underlying ground-truth function [65], which might be the case here, regarding the maximum accuracies of 80% on the full dataset.

Although steady performance increases over iterations for all models were observed, the non-neural models show clearer learning trends than the NN (head) models (see figures 13 and 14). This can be explained by the better overall performance of non-neural ML models like boosted trees for tabular data that is widely observed, implying better model performance is achievable with less labeling effort in general. Furthermore, NN models need hyperparameter optimization that is more challenging in the context of few data [66, 67].

The basis for these interpretations were learning curves over AL iterations

and the computed AULC values. One drawback of using AULC is its limited interpretability. For instance, the same AULC value can result from two curves with different shapes and end qualities [68]. Therefore, it is important to look at learning curves likewise as it was done in this study and never interpret AULC values alone. Second, AULC values cover the full learning curve until the last iteration, which equals full dataset coverage here. As AL is designed to limit labeling effort, an unlabeled dataset is typically not fully labeled in real-world AL experiments. Furthermore, the largest accuracy increase and therefore, model improvement, can be observed in the first few iterations, whereas a plateau typically arises towards the last iterations. This is why in future experiments, using a partial AULC covering only a fraction of iterations (e.g. AULC₅₀ for the first 50% of iterations) should be considered.

Further limitations of this study concern dataset size and model choice. First, the datasets from Landrum et al. [26] consist of less than 1000 samples each, which might reduce the ability to generalize the results of this study to larger datasets. Second, only tabular ML models were considered in this work, although molecules can be naturally represented by graphs with atoms as nodes and bonds as edges. Subsequently, this representation can be leveraged to train graph neural networks (GNNs) [69].

5.3 Future work directions

In this study, predicting bioactivity of molecular compounds was formulated as classification task, defining if a compound is active (1) or inactive (0) against a specific biological target. K_i values were measured for all compounds in the respective datasets, CA-IV, hERG and OX1R, representing binding affinity of a compound against the target. This numerical K_i value could be directly used as target variable in a regression problem, which is a problem setup left for future work. Based on this, AL for regression problems [70] could be investigated.

Furthermore, TabPFN and CatBoost as tabular classifiers have shown elevated learning trends in the AL experiments, but are limited to categorical and numerical features and cannot be directly applied to the wide-spread usage of computer vision. To investigate the concordance of this study’s findings with data from computer vision tasks (e.g. phenotypic profiling of small molecules in drug discovery [71, 72]), future work could focus on combining these tabular classifiers with feature embeddings extracted from convolutional NNs or vision transformers.

6 Conclusion

This study aimed to investigate how model choice and data representation influence UQ and AL effectiveness with molecular datasets. Therefore, three different datasets were chosen, measuring bioactivity of compounds against the biological targets CA-IV, hERG and OX1R. The datasets were first used to evaluate different classification models, being a baseline NN, CatBoost and TabPFN, towards calibration of predicted confidence scores and model uncertainty. For each dataset, model performance and reliability was assessed using several representation types for molecular data. These representations were (1) RDKit descriptors, (2) Morgan fingerprints and (3) CheMeleon embeddings.

Second, AL was performed with the OX1R dataset and RDKit descriptors as feature representation. By assessing balanced accuracy after each learning cycle and calculating the AULC across the complete learning curve, uncertainty-based query methods were compared to the random acquisition strategy, revealing how uncertainty quantification can leverage learning efficiency.

It can be concluded that predictive performance is less dependent on model type and feature representation than calibration quality, as shown in calibration curves, ECE and Brier score comparisons. All investigated model architectures tend to predict with overconfidence, especially the NN used in this study. Predictive entropy and ensemble disagreement provide additional uncertainty estimates, capturing total and epistemic uncertainty, respectively.

In AL, uncertainty-based query methods were able to outperform random acquisition in several cases, whereas the combination of uncertainty with aspects of data diversity for acquisition performs even better, bridging exploration and exploitation strategies. However, reliable uncertainty estimation is a prerequisite for efficient AL with molecular data.

These results contribute to a better understanding of the reliability in molecular prediction that is influenced by aspects like model architecture and data representation. Thereby, experimental costs in safety-critical applications such as bioactivity prediction can be reduced and data efficiency can be improved by implementing multiple learning cycles. While the study provides a systematic evaluation, the conclusions are limited to the investigated datasets and may not fully generalize to prospective drug discovery settings.

Further limitations include the dominance of specific scaffolds in one of the datasets, which might have induced overfitting of models. Second, feature representations could be extended to different lengths or combinations in future work. Third, the ECE and Brier score used as calibration measures here both come with disadvantages, that were not directly addressed, e.g. by decomposing Brier score. For AL, AULC was used, which should be replaced by a partial AULC measure in future studies in order to cover learning improvements more meaningfully.

By systematically analyzing predictive uncertainty and its role in active learning, this work advances the understanding of predictive uncertainty in bioactivity modeling and demonstrates its potential to enable more reliable and data-efficient molecular prediction.

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A Appendix

For BI and BALD acquisition in active learning, different values of n were evaluated. The value represents the number of samples that are drawn to span the posterior distribution for computation of BALD and BI (see Section 3.3). The resulting learning curves for CatBoost and the NN model are shown in Figure 17 for BI and Figure 18 for BALD acquisition.

As the curves mostly overlap, $n = 30$ was set as default value in the end.

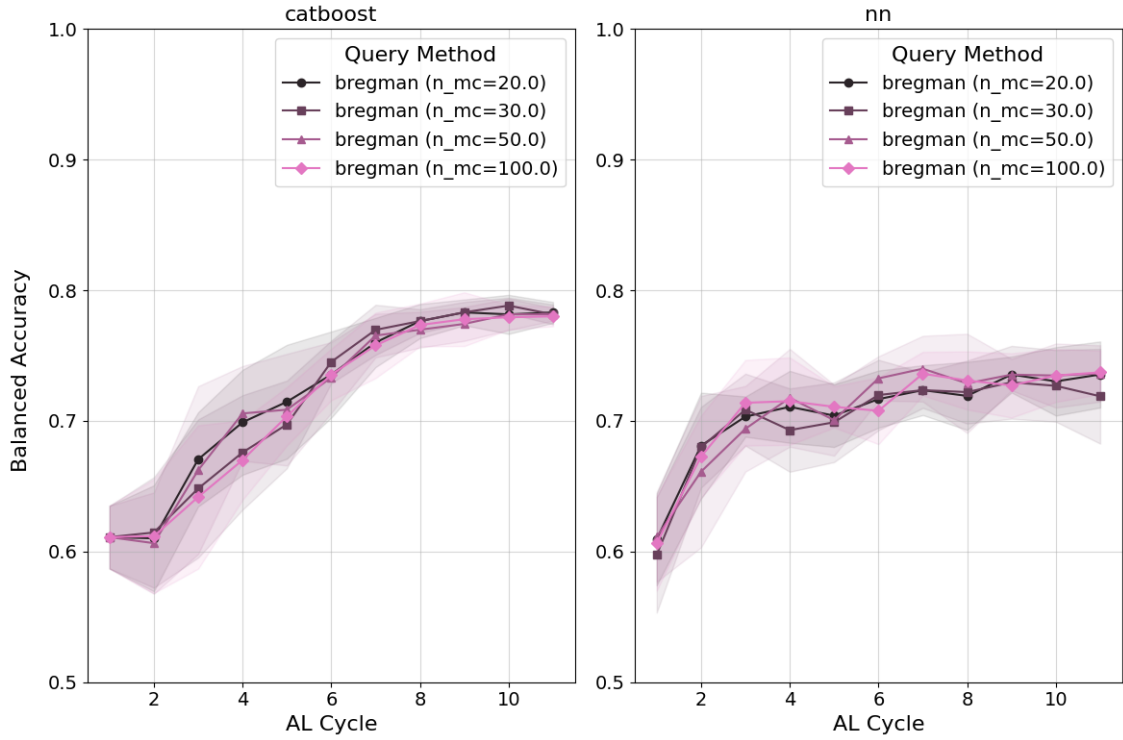


Figure 17: Learning curves for varying n in BI acquisition
For CatBoost (left) and NN (right), different numbers of samples spanning the posterior distribution (n_{mc}) were evaluated.

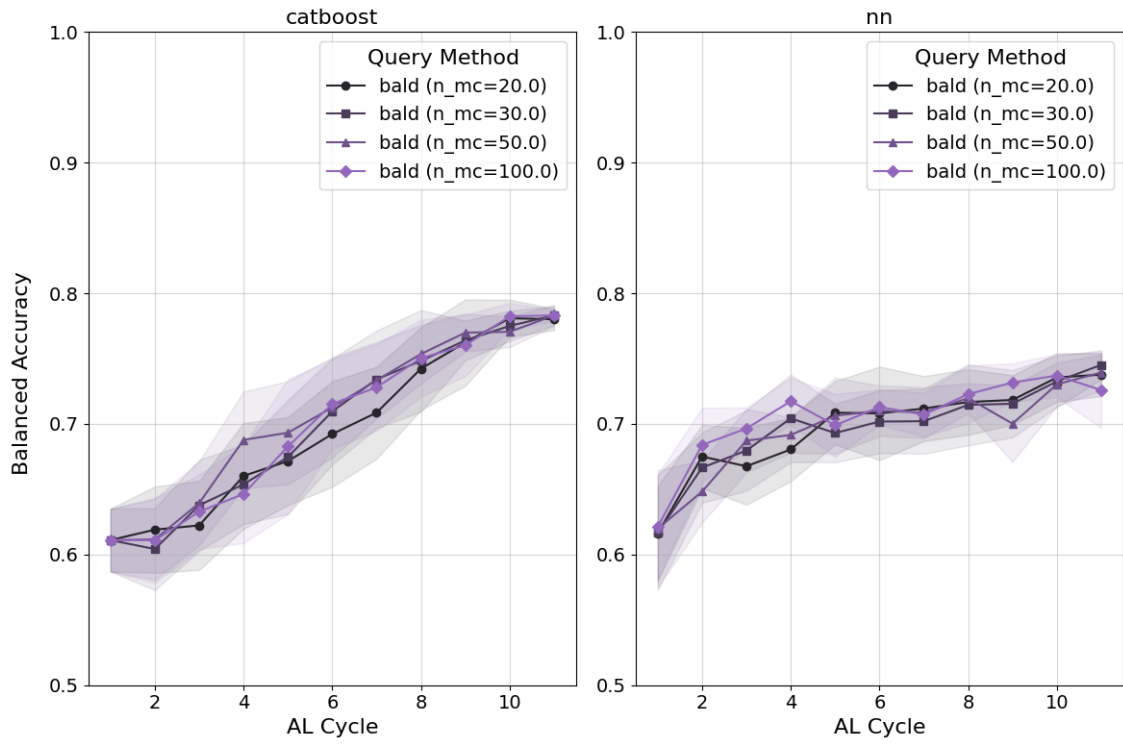


Figure 18: Learning curves for varying n in BALD acquisition
For CatBoost (left) and NN (right), different numbers of samples spanning the posterior distribution (n_{mc}) were evaluated.

Eidesstattliche Erklärung

Ich versichere, dass ich die vorliegende Arbeit ohne fremde Hilfe und ohne Benutzung anderer als der angegebenen Quellen angefertigt habe, und dass die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegen hat. Alle Ausführungen der Arbeit, die wörtlich oder sinngemäß übernommen wurden, sind als solche gekennzeichnet.